

DIGITAL FORENSICS INCIDENT RESPONSE (DFIR) REPORT

Case Reference: M57-Jean Data Exfiltration

Date of Investigation: 2026-06-12 17:30 IST

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Target Media: nps-2008-jean.E01 (Windows XP Endpoint)

1. EXECUTIVE SUMMARY On July 20, 2008, a data exfiltration incident occurred on the corporate endpoint assigned to the user "Jean." A forensic analysis of the system's disk image was conducted to determine the scope and vector of the data loss. The investigation concluded that the user was the victim of a targeted Business Email Compromise (BEC) and spear-phishing attack. An external threat actor successfully spoofed the identity of an internal executive to manipulate the user into compiling and emailing highly sensitive Personally Identifiable Information (PII), including employee salaries and Social Security Numbers. There is no evidence of malware infection or internal collusion.

2. SCOPE AND OBJECTIVES

- Image Acquisition: EnCase format (.E01) disk image of the compromised workstation.
- Objectives:
 1. Identify the exact method of data exfiltration.
 2. Determine the timeline of events leading up to the breach.
 3. Assess the involvement of internal vs. external threat actors.

3. FORENSIC METHODOLOGY & TOOLS The analysis was conducted using Autopsy. The disk image was processed using the following ingest modules:

- **Recent Activity:** To extract web history, recent documents, and registry artifacts.
- **Keyword Search:** To index and locate specific strings related to the compromised data (e.g., "m57biz.xls").
- **Email Parser:** To reconstruct local email communication files (Thunderbird/Outlook).

4. TIMELINE OF KEY EVENTS

- **July 19, 2008 (16:39 PDT):** The threat actor utilizes an external hosting provider (Dreamhost) to forge the initial email headers.
- **July 20, 2008 (05:09 IST):** User "Jean" receives the initial grooming email. The sender display name is spoofed to read alison@m57.biz, requesting a compilation of employee PII.

- **July 20, 2008 (06:58:03 IST):** User complies with the request and creates/modifies the file m57biz.xls on the local Desktop.
- **July 20, 2008 (06:58:00 IST):** User replies to a secondary, urgent email from the attacker, attaching m57biz.xls. The reply is routed to the attacker-controlled address tuckgorge@gmail.com. *(Note: The 3-second discrepancy between local file creation and email submission is standard latency within forensic parameters).*

5. TECHNICAL FINDINGS 5.1. Social Engineering Lure (Email Spoofing) The initial request for data appeared to originate from a corporate executive. However, raw SMTP header analysis revealed the email bypassed internal routing completely.

- **Return-Path:** <simsong@xy.dreamhostps.com>
- **Originating Server:** smarty.dreamhost.com

5.2. Payload Creation File system metadata confirms the compilation of the requested data immediately following the spoofed request.

- **File Path:** Documents and Settings/Jean/Desktop/m57biz.xls
- **Creation Timestamp:** 2008-07-20 06:58:03 IST

5.3. Exfiltration Vector The data was exfiltrated via standard SMTP traffic. The attacker utilized a mailto: parameter in the spoofed email to ensure the victim's reply was routed to an external inbox rather than the actual executive's inbox.

- **Destination Address:** tuckgorge@gmail.com

6. INSIDER THREAT ASSESSMENT A comprehensive review of the Communications Visualization timeline and system access logs isolated the communication strictly between the victim and the external threat actor. No lateral movement or secondary unauthorized logins were detected. The incident is classified exclusively as an external BEC attack.

7. REMEDIATION & RECOMMENDATIONS To prevent future occurrences of this attack vector, the following enterprise-wide controls are recommended:

1. **Implement Email Authentication:** Enforce strict SPF, DKIM, and DMARC policies on the corporate domain to reject externally originating emails that attempt to spoof internal addresses.
2. **External Sender Warnings:** Configure the corporate mail gateway to append a visual warning banner to all emails originating from outside the organization.
3. **Data Loss Prevention (DLP):** Deploy endpoint and network DLP solutions to detect and block the outbound transmission of structured PII (such as SSNs).